

Global Engineering Documentation Estate

Minimum controlled documents required to build, verify, launch and scale a global public-markets platform

DOCUMENT ID	PLT-ENG-DOC-001	VERSION	2.0
STATUS	Proposed baseline — for engineering governance	ARCHITECTURE SOURCE	PLT-ENG-ARCH-003 v3.0
GLOBAL PLATFORM	Multi-jurisdiction • Multi-market • Multi-asset	INITIAL MARKET WEDGE	UK gilts + US Treasuries only

GLOBAL SCOPE CORRECTION

Talvrin is a global public-markets platform. UK gilts and US Treasuries are the first implementation wedge only. No core schema, service, API, policy model, calculation framework, identifier strategy, data-rights model or operating control may assume that the platform is permanently limited to the United Kingdom or United States.

1. Governing Documentation Rule

The engineering documentation estate remains deliberately lean at 14 controlled documents. Global scope is achieved by designing the core contracts to be jurisdiction-neutral and asset-neutral, then expressing market-specific differences as controlled annexes and configuration packs inside the governing specifications — not by multiplying standalone documents for every country, exchange or asset class.

- Country-specific rules belong in versioned jurisdiction packs governed by PLT-POL-001.
- Market-data source differences belong in source profiles and market packs governed by PLT-DATA-002 and PLT-RIGHTS-001.
- Asset-class formulas and conventions belong in asset methodology packs governed by PLT-FIN-001.
- Currencies, time zones, calendars, locales and market identifiers are data/configuration, never hard-coded country assumptions.
- A new top-level document may be created only where an ADR proves that a material build contract cannot be governed within this estate.

Global-by-design rule

The core platform must support additional jurisdictions, exchanges, data vendors, languages, currencies and public-market asset classes without database re-platforming or replacement of the principal domain contracts. Expansion should add governed packs and adapters, not fork the platform.

2. Required Global Engineering Document Set

These are the only Talvrin-specific narrative engineering documents required for the platform baseline. PLT-ENG-ARCH-003 is complete; the remaining 13 documents are the controlled build specifications.

#	DOCUMENT ID	DOCUMENT	PURPOSE / GLOBAL BUILD CONTRACT	STATUS
1	PLT-PRD-001	Global Product Requirements & Functional Specification	Defines the global product contract: launch scope, target operating model, user journeys, functional requirements, market/jurisdiction expansion rules, localisation requirements, exclusions, acceptance criteria and tier behaviour.	TO PREPARE
2	PLT-ENG-ARCH-003	Backend Architecture & Engineering System Specification	Governs the launch and target architecture, global-by-design doctrine, persistence, failure semantics, NFRs, scaling triggers, build sequence and architectural gates.	COMPLETE
3	PLT-DATA-001	Canonical Global Data Model & Database Specification	Defines jurisdiction-neutral canonical schemas and invariants for instruments, issuers, venues, currencies, identifiers, source observations, accepted facts, evidence, calculations, users, rules, alerts, rights and audit entities.	TO PREPARE
4	PLT-DATA-002	Global Market Data Ingestion, Normalisation & Reconciliation Specification	Defines the connector framework, global source onboarding, market/asset adapters, validation, transformation, identity resolution, source precedence, reconciliation, corrections, revocations, freshness and degradation.	TO PREPARE
5	PLT-RIGHTS-001	Global Data Rights, Entitlements & Source Governance Specification	Converts vendor, exchange, index-provider, issuer and public-source rights into enforceable retrieve/store/cache/display/API/embed/export/API/redistribution, territorial, retention and purge controls.	TO PREPARE
6	PLT-FIN-001	Global Financial Calculation, Market Convention & Methodology Specification	Defines the calculation framework and versioned asset/market methodology packs: formulas, quotation conventions, calendars, settlement, day count, rounding, precision, authoritative sources, golden values and methodology governance.	TO PREPARE
7	PLT-MON-001	Global Monitoring, Rules & Alerting Specification	Defines market-agnostic monitoring: rules, thresholds, crossings, state changes, scheduled events, hysteresis, debounce, late/stale data, corrections, coverage, delivery, time zones and auditability.	TO PREPARE
8	PLT-EVID-001	Evidence, Provenance & Global Document Intelligence Specification	Defines source → observation → accepted fact → calculation → evidence bundle → citation across jurisdictions, plus document acquisition, parsing, language handling, indexing and source-location rules.	TO PREPARE

#	DOCUMENT ID	DOCUMENT	PURPOSE / GLOBAL BUILD CONTRACT	STATUS
9	PLT-AI-001	AI Gateway, Retrieval & Model Safety Specification	Defines provider-neutral AI orchestration, evidence-constrained retrieval, multilingual handling, prompt/model versioning, numeric grounding, response types, injection defence, evaluation gates and degradation.	TO PREPARE
10	PLT-API-001	Global Application API & Integration Contract	Defines stable web/mobile/enterprise contracts: resources, commands, schemas, auth, pagination, errors, idempotency, async jobs, locale/currency/time-zone semantics, versioning and entitlement behaviour.	TO PREPARE
11	PLT-POL-001	Global Jurisdiction, Capability & Regulatory Policy Specification	Converts the operating mandate into executable jurisdiction packs, regulatory-perimeter rules, prohibited capabilities, recommendation-safe patterns, legal-entity/country feature gating and policy decisions.	TO PREPARE
12	PLT-SEC-001	Security, Privacy, Data Residency & Cross-Border Access Specification	Defines identity, MFA/passkeys, RBAC/ABAC, tenancy, encryption, secrets, PII separation, residency, international data transfers, retention/erasure, privileged access and audit security.	TO PREPARE
13	PLT-QE-001	Global Quality Engineering, Verification & Release Standard	Defines verification across market packs: unit/integration/contract, bitemporal, financial golden, monitoring, rights, AI, policy/perimeter, localisation, tenancy, load, chaos, security and release gates.	TO PREPARE
14	PLT-OPS-001	Global Platform Operations, Deployment & Production Readiness Specification	Defines environments, CI/CD, IaC, observability, SLOs, source health, regional deployment strategy, monitoring-of-monitoring, backups, restore, DR, incidents, kill switches and readiness.	TO PREPARE

3. Controlled Build Sequence

The sequence is dependency-led. Global capability is not deferred until after a UK/US build; rather, the core contracts are global from the outset while the first executable market packs are UK gilts and US Treasuries.

ORDER / GATE	DOCUMENT	WHY NOW	GLOBAL PRINCIPLE
1 — FOUNDATION	PLT-PRD-001	Locks global product scope and expansion rules.	Core contract first; market/jurisdiction packs extend it without forking.
2 — DATA MODEL	PLT-DATA-001	Creates canonical global objects and invariants.	Core contract first; market/jurisdiction packs extend it without forking.

ORDER / GATE	DOCUMENT	WHY NOW	GLOBAL PRINCIPLE
3 — RIGHTS	PLT-RIGHTS-001	Defines source, territorial and processing permissions before ingestion.	Core contract first; market/jurisdiction packs extend it without forking.
4 — MARKET DATA	PLT-DATA-002	Implements the global connector/adaptor and reconciliation framework.	Core contract first; market/jurisdiction packs extend it without forking.
5 — FINANCIAL LOGIC	PLT-FIN-001	Pins calculation framework plus first asset/market methodology packs.	Core contract first; market/jurisdiction packs extend it without forking.
6 — POLICY	PLT-POL-001	Defines jurisdiction packs and capability boundaries.	Core contract first; market/jurisdiction packs extend it without forking.
7 — MONITORING	PLT-MON-001	Specifies the recurring-value engine independently of one market.	Core contract first; market/jurisdiction packs extend it without forking.
8 — EVIDENCE	PLT-EVID-001	Defines global provenance and document-intelligence chain.	Core contract first; market/jurisdiction packs extend it without forking.
9 — AI	PLT-AI-001	Layers controlled model behaviour over trusted evidence.	Core contract first; market/jurisdiction packs extend it without forking.
10 — API	PLT-API-001	Freezes global client/backend integration semantics.	Core contract first; market/jurisdiction packs extend it without forking.
11 — SECURITY	PLT-SEC-001	Hardens cross-border identity, privacy, residency and access.	Core contract first; market/jurisdiction packs extend it without forking.
12 — QUALITY	PLT-QE-001	Proves every market/jurisdiction pack against common release gates.	Core contract first; market/jurisdiction packs extend it without forking.
13 — OPERATIONS	PLT-OPS-001	Defines production operation and later regional expansion.	Core contract first; market/jurisdiction packs extend it without forking.

4. Global Expansion Is Governed by Packs, Not New Top-Level Documents

To prevent document proliferation while preserving jurisdictional precision, Talvrin uses controlled packs and annexes inside the 14-document estate.

PACK	GOVERNED WITHIN	CONTENT
Jurisdiction Pack	PLT-POL-001	Country/territory regulatory perimeter, legal entity, feature permissions/prohibitions, financial-promotion rules, required disclosures and counsel approval.
Market / Source Pack	PLT-DATA-002 + PLT-RIGHTS-001	Exchange/vendor/source onboarding, identifiers, data latency, redistribution, territorial rights, market hours, source precedence and degradation.
Asset / Methodology Pack	PLT-FIN-001	Asset-specific calculation catalogue, quotation and settlement conventions, day-count methods, calendars, formula references and golden values.
Locale / Experience Pack	PLT-PRD-001 + PLT-API-001	Language, number/date formats, currency display, time-zone handling, terminology and locale-specific user experience.

PACK	GOVERNED WITHIN	CONTENT
Deployment / Residency Pack	PLT-SEC-001 + PLT-OPS-001	Approved processing region, residency, transfer mechanism, backup/DR region and operational constraints.

5. Documents That Must Not Be Created Separately

The following subjects are necessary, but they belong inside the controlled estate. Country-by-country or asset-by-asset standalone documents would duplicate authority and create drift.

SUBJECT	CONTROLLED WITHIN / DECISION
Country / jurisdiction specifications	Jurisdiction packs inside PLT-POL-001
Exchange-specific architecture	Market/source packs inside PLT-DATA-002 + PLT-RIGHTS-001
Country-specific database schemas	Prohibited; use PLT-DATA-001 canonical model + configuration
Individual asset-class architecture documents	Asset/methodology packs inside PLT-FIN-001 and PLT-DATA-002
Instrument Master / bitemporal model	PLT-DATA-001
Source Registry / vendor exit / purge	PLT-RIGHTS-001
Evidence Ledger	PLT-EVID-001
Calculation API	PLT-API-001 + PLT-FIN-001
Alert API	PLT-API-001 + PLT-MON-001
RAG / prompts / model evaluations	PLT-AI-001 + PLT-EVID-001 + PLT-QE-001
Authentication / database security / cross-border access	PLT-SEC-001
CI/CD / infrastructure / observability / DR	PLT-OPS-001
Separate localisation document	PLT-PRD-001 + PLT-API-001 + PLT-QE-001
Separate NFR document	PLT-ENG-ARCH-003 + PLT-OPS-001 + PLT-QE-001

6. Engineering Governance Decision

LOCK THE GLOBAL TALVRIN ENGINEERING ESTATE AT 14 DOCUMENTS — New countries, exchanges, vendors, languages, currencies and asset classes are governed through versioned packs and adapters inside these specifications. A new top-level document requires an ADR proving a genuinely new architectural authority boundary.